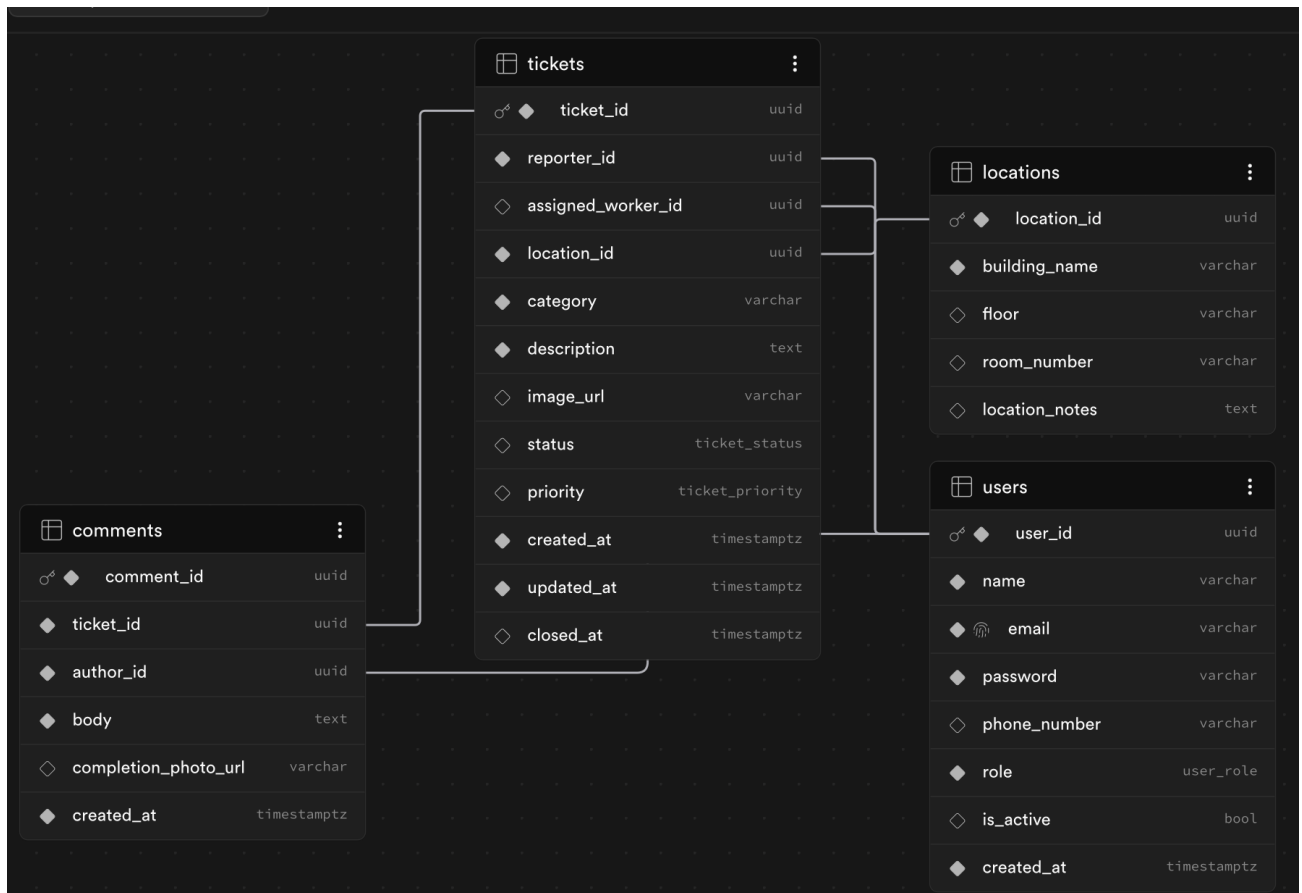


Database Schema Documentation

Overview

The CampusCare system uses a PostgreSQL relational database. The schema is designed with strong referential integrity, using foreign keys, custom ENUM types, and indices for performance.



1. Custom Types (ENUMs)

To ensure data consistency, the following custom types are used in the database:

Type Name: Allowed Values

user_role: community_member, worker, facility_manager, admin

ticket_status: pending, assigned, in_progress, resolved, denied

ticket_priority: low, normal, high, critical

2. Table Definitions

2.1 Users Table (public.users)

Stores account information and access levels.

Column Name : Data Type : Constraints : Description

user_id : uuid : PK, DEFAULT gen_random_uuid() : Unique user ID

name : varchar(100) : NOT NULL : Full name

email : varchar(150) : NOT NULL, UNIQUE : University email

password : varchar(255) : NOT NULL : Bcrypt-hashed password

role : user_role : NOT NULL : User permissions role

is_active : boolean : DEFAULT true : Activation status

2.2 Locations Table (public.locations)

Stores unique locations reported in tickets.

Column Name : Data Type : Constraints : Description

location_id : uuid : PK, DEFAULT gen_random_uuid() : Unique location ID

building_name : varchar(100) : NOT NULL : Building name

floor : varchar(20) : : Floor level

room_number : varchar(50) : : Room identifier

2.3 Tickets Table (public.tickets)

The core repository for all reported facility issues.

Column Name : Data Type : Constraints : Description

ticket_id : uuid : PK : Unique ticket ID

reporter_id : uuid : FK (users) : The reporter's ID

assigned_worker_id : uuid : FK (users) : Assigned maintenance worker

location_id : uuid : FK (locations) : Linked location details.

Category : varchar(50) : NOT NULL : Issue type (e.g., Plumbing)

Status : ticket_status : DEFAULT 'pending' : Lifecycle state

Priority : ticket_priority : DEFAULT 'medium' : Urgency level

2.4 Comments Table (public.comments)

Stores progress logs and photographic proof of completion.

Column Name : Data Type : Constraints : Description

comment_id : uuid : PK : Unique comment ID

ticket_id : uuid : FK (tickets) : Associated ticket

author_id : uuid : FK (users) : Author of the note

body : text : NOT NULL : Text content of the update

completion_photo_url : varchar(500) : : Image URL for work proof

Migrations:

```
-- CampusCare - Initial Schema
-- Run this in your PostgreSQL database (e.g., Supabase SQL Editor).

-- 1. Create ENUM types
create type user_role as enum ('community_member', 'worker', 'facility_manager',
'admin');
create type ticket_status as enum ('pending', 'assigned', 'in_progress', 'resolved',
'denied');
create type ticket_priority as enum ('low', 'normal', 'high', 'critical');

-- 2. Users Table
create table public.users (
  user_id      uuid primary key default gen_random_uuid(),
  name         varchar(100) not null,
  email        varchar(150) not null unique,
  password     varchar(255) not null,
  phone_number varchar(20),
  role         user_role not null,
  is_active    boolean default true,
  created_at   timestampz not null default now()
);

-- 3. Locations Table
create table public.locations (
  location_id  uuid primary key default gen_random_uuid(),
  building_name varchar(100) not null,
  floor        varchar(20),
  room_number  varchar(50)
);

-- 4. Tickets Table
create table public.tickets (
  ticket_id      uuid primary key default gen_random_uuid(),
  reporter_id    uuid not null references public.users(user_id) on delete
cascade,
  assigned_worker_id uuid references public.users(user_id),
  location_id    uuid not null references public.locations(location_id),
  category       varchar(50) not null,
  description    text not null,
  image_url      varchar(500),
  status         ticket_status default 'pending',
  priority       ticket_priority default 'normal',
  created_at     timestampz not null default now(),
  updated_at     timestampz not null default now(),
  closed_at      timestampz
);
```

```

-- 5. Comments Table
create table public.comments (
  comment_id      uuid primary key default gen_random_uuid(),
  ticket_id       uuid not null references public.tickets(ticket_id) on delete
  cascade,
  author_id       uuid not null references public.users(user_id) on delete
  cascade,
  body            text not null,
  completion_photo_url varchar(500),
  created_at      timestampz not null default now()
);

-- 6. Indices for Performance
create index idx_tickets_reporter on public.tickets(reporter_id);
create index idx_tickets_worker on public.tickets(assigned_worker_id);
create index idx_tickets_status on public.tickets(status);
create index idx_comments_ticket on public.comments(ticket_id);

```

sample:

